

Physical Card Delivery – Technical Design Overview

Overview

This document describes the technical design for the Physical Card Delivery Status Processing System.

The solution is responsible for receiving delivery status files from an external MDP provider, validating and processing them using Spring Batch, updating card delivery status in the CRM backend, and managing file lifecycle across multiple SFTP locations.

High-Level Architecture

The solution is implemented using Apache Camel as a central orchestration layer and consists of two independent processing routes:

Route 1 – MDP File Handoff

Responsible for securely transferring delivery status files from the MDP SFTP server to the ONEBANK SFTP server.

Route 2 – Card Delivery Status Processing

Responsible for validating, staging, processing, and updating card delivery status in the CRM backend using Spring Batch.

The solution guarantees:

- Exactly-once file processing.
- Fault isolation and recovery.
- End-to-end traceability.

Main Components

Component	Responsibility
Apache Camel	Orchestration is responsible for file polling, routing, validation, error handling, and integration between systems.
MDP SFTP	External provider SFTP server hosting card delivery status files
ONEBANK SFTP	Internal SFTP server used as the bank's integration entry point
Spring Batch	Batch processing engine responsible for parsing files and updating CRM backend

MongoDB	Store idempotent keys, job execution status, job parameters and read locks.
File System (Local Staging)	Temporary local directory used for batch processing
CRM Backend	Target system for updating physical card delivery status.

Processing Flow Overview

Route 1 – MDP File Handoff

- ❖ Apache Camel polls **/home/mdp_delivery_card_status** on MDP SFTP.
- ❖ Files are protected using:
 - ReadLock to prevent partial reads and multi-pod safe polling.
 - Idempotent repository (MongoDB) with eager=false
- ❖ Files are uploaded to ONEBANK SFTP directory **/home/mdp_delivery_card_status**

Failure Handling

- In case of transfer failure, the transaction is rolled back.
- The idempotent key is not persisted.
- The file is retried automatically on the next poll cycle.
- Failure is logged to Monitoring.

Route 2 – Card Delivery Status Processing

- ❖ Apache Camel polls **/home/mdp_delivery_card_status** on ONEBANK SFTP.
- ❖ Files are protected using:
 - ReadLock to prevent partial reads and multi-pod safe polling
 - Idempotent repository (MongoDB) with eager=false
- ❖ File name and extension are validated.

Validation Failure

- File is moved to /home/failed
- Validation error is logged

Validation Success

- File is staged to local directory /home/java/
- Job execution status is checked in MongoDB

Case A — Job Already Completed

- File is moved to /home/processed
- Event is logged

Case B — Job Not Processed

- Spring Batch job is launched
- Batch engine:
 - Reads file from local staging
 - Processes records in chunks (example: 100).
 - Updates CRM backend in bulk.

Batch Failure

- Transaction is rolled back
- Idempotent key is not stored
- File remains eligible for retry
- Failure is logged

Batch Success

- File is moved to /home/processed
- Success is logged
- Local staged file is removed